

NEWSLABS TRIGGER · INTERNAL

SanDisk (SNDK) Momentum Episodes & Daily Digest

A → Z product walkthrough + engineering deep dive · Full trading-day timeline

This brief explains how Trigger detects and notifies real-time price momentum using **SNDK (SanDisk)** as the working example, how **Daily Digest** differs from episodes, and how both channels behave on a single trading day. Written for leadership and engineering.

Example ticker	SNDK (SanDisk)
Audience	Boss walkthrough + engineering deep dive (English)
Part A	Momentum episode state machine, alerts, windows, re-arm, races
Part B	Daily Digest OPEN / MIDDAY / CLOSE — independent of episodes
Part C	One full SNDK trading day — both channels interleaved
Poll cadence	~60 seconds (configurable via MOMENTUM_POLL_MS)
Classification	Confidential · Internal product brief

One sentence (episodes): Trigger watches SNDK across many time windows; when a real short-horizon move crosses a threshold it opens **one episode**, freezes how that move is measured, and only notifies on meaningful milestones — not every price tick.

One sentence (digest): Daily Digest is a separate scheduled channel — open / midday / close snapshots vs previous session close for equities and indexes — and never creates or ends episodes.

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Leadership path: START → HOLD → ACCEL → WEAKEN → STRONG FADE → RE-ACCEL → EXPIRE/REVERSE. Two spam guarantees: **shortest-window START** and **global direction re-arm**. Digest is a third, calendar-driven channel.

Part A — Momentum Episodes (SNDK A→Z)

A1. Product definition & design principles

A **momentum episode** is one active directional story for one ticker (e.g. SNDK UP). It is not a daily P&L label and not a permanent “stock is high” flag. It represents **active expansion or fade of a move** that just crossed a configured threshold.

- **One active episode per ticker** — never two concurrent STARTs for SNDK.
- **One frozen reference** — all episode math uses the same reference price for the whole life.
- **Milestone alerts only** — START (after research), +2pp extension, 60% giveback, re-accel, true reverse.
- **No re-packaging** — after expire, a longer window cannot re-alert the same elevated move.
- **Research-backed START** — push body is the likely driver (Perplexity), not generic noise.

A2. Poll loop & eligible windows

Default poll interval is ~60 seconds (not every second). Engineering: MOMENTUM_POLL_MS. On each poll for SNDK:

Step	Action	Detail
1	Fetch price	Yahoo live quote + 1-minute candles
2	Compute returns	5m, 15m, 1h, 3h, 8h, 24h, day (vs prev close), ...
3	Threshold compare	Each window has its own move % thr (settings / Supabase)
4a	No ACTIVE	Re-arm check → hits → shortest window START
4b	ACTIVE	Extremes from frozen ref → reverse / expire / giveback / accel
5	Terminal	EXPIRED / ENDED / REVERSED / MARKET CLOSE immutable

UI blink: rolling-return cards blink only when that window's **own configured threshold** is crossed (not a fixed cosmetic 3%).
Episode-eligible: ≤24h rolling + day (1D). Multi-day (1w, 10d, 1M) are diagnostic only and never open an episode. Code: `isEpisodeEligibleWindow`.

A3. Policy defaults

Rule	Default	Notes / env
Poll interval	~60s	MOMENTUM_POLL_MS
5m thr	≥ 3%	Typical equity START window
15m / 10m	≥ 3%	Configurable per asset class
60m	≥ 5%	
3h	≥ 6%	
Day vs prev close	≥ 5%	Regular session card
Acceleration alert	+2 pp	MOMENTUM_ACCEL_POINTS
Holding → Weakening	≥ 25% giveback	State only — no push
Weakening → Holding	≤ 20% giveback	Hysteresis
Strong giveback push	≥ 60% giveback	Once per fade cycle
Expiry	60 min	Without ≥0.5 pp new extreme
Re-arm buffer	thr – 1 pp	MOMENTUM_REARM_BUFFER_PP
START push max age	~5 min	MOMENTUM_START_PUSH_MAX_AGE_MS

Thresholds are per asset class (equity / crypto / forex / commodity / index) and editable in the dashboard Settings rail. Blank or 0 = window off (never START).

A4. Frozen source of truth

At START the engine freezes measurement identity. Every later % uses:

$$((\text{livePrice} - \text{frozenReferencePrice}) / \text{frozenReferencePrice}) \times 100$$

Field	Meaning	Example
detectedWindow	Winning short window	5m
referencePrice	Price at lookback	\$100.00
referenceTime	Bar time of reference	10:00 – 5m
triggerMovePct	Move at START	+3.20%
exactMinutes / Label	True elapsed span	5 minutes
peakMovePercent	Best extension	+8.0%
lastNotifiedEpisodeMovePct	Last push baseline	+5.2%

We never swap mid-episode to day return or 24h return for the same alert story. Engineering: createEpisodeFromHit freezes ref; updateExtremes only mutates live/peak/trough from that ref.

A5. START rules & multi-window selection

A5.1 Simple start

10:00 — SNDK was \$100 five minutes ago; live **\$103.20** → 5m = **+3.2%** (thr 3%).

- One episode: **UP**, detectedWindow = **5m**; freeze ref \$100, move +3.2%
- State: STARTED (brief dwell) → HOLDING
- Raw START does **not** auto-push; Perplexity research runs first
- Push only if still ACTIVE + same episodeld + trigger age ≤ ~5 min
- Example title: **SNDK is surging** ↑ **+3.2% in 5 minutes**; body = likely driver

A5.2 Many windows qualify — shortest wins

Window	Move	Thr	Qualifies?	Selected?
5m	+3.2%	3%	Yes	YES — shortest
15m	+4.0%	3%	Yes	No
1h	+5.5%	5%	Yes	No
3h	+7.0%	6%	Yes	No
24h	+8.0%	5% if on	Maybe	No

Only one episode. Freeze 5m ref and +3.2%. Longer co-qualifiers are logged as also-qualified and ignored — no second START. Engineering: pickShortestStartHit.

A5.3 Direction from shortest window (not strongest |move|)

If 24h is **−8%** (stale) and 5m is **+3.5%** (fresh), shortest wins → **UP 5m**. A large absolute long-horizon move cannot override a fresh short move. Rule: **shortest qualifying window overall sets direction and freezes that window's reference**.

A5.4 While ACTIVE

A newly qualifying shorter window does **not** open another START. It only matters if price extension vs the frozen reference hits +2pp acceleration rules.

A6. State map, transitions, giveback math

STARTED → HOLDING → (ACCELERATING and/or WEAKENING) → STRONGLY_WEAKENING → RE_ACCELERATING → ... or REVERSED / EXPIRED / MARKET CLOSE / ENDED at any time.

State	Meaning	User push?
STARTED	Just opened; brief dwell	After research if fresh
HOLDING	Near peak; giveback < 25%	No
ACCELERATING	≥ +2 pp beyond last notified	Yes
WEAKENING	≥ 25% of peak move given back	No
STRONGLY_WEAKENING	≥ 60% giveback this cycle	Yes (strong giveback)
RE_ACCELERATING	≥ +2 pp from recovery anchor	Yes
EXPIRED	60 min without ≥0.5 pp extreme	No (silent)
REVERSED	Through ref + opposite thr	Yes (reverse research)
ENDED	Manual end	No
CLOSED_AT_MARKET_CLOSE	US equity session closed	No

Terminal states are immutable. Same episode never re-enters Weakening/Accel after EXPIRED/ENDED/REVERSED/MARKET CLOSE. Engineering: terminal check returns early; disarm then re-arm gate.

Giveback (UP): (peakPrice – live) / (peakPrice – reference). Ref \$100, peak \$110 (+10%), live \$104 (+4%) → giveback = 60%.
Event name for ≥60% fade is **MOMENTUM_STRONG_WEAKENING** (not STRONG_REVERSAL). True REVERSAL is a different path.

A7. Alerts — when we push and what we say

A) START (after research)

When: new episode + research done + same episode ACTIVE + trigger age $\leq$ ~5 min. Title e.g. *SNDK is surging* $\uparrow$ +3.2% in 5 minutes. Body = likely driver only. Late research: save research, suppress push (START_PUSH_MAX_AGE_MS).

B) Acceleration (+2 pp)

When: last notified +3.2% $\rightarrow$ now +5.4% (+2.2 pp) on frozen ref. Tiny +0.03 pp never alerts.

C) Strong giveback ($\geq 60\%$)

Title e.g. *SNDK has given back 60% of its surge*. Sets recovery anchor at remaining move. One push per fade cycle; cycle can reset after re-accel.

D) Re-acceleration

After strong fade, recover $\geq$ +2 pp from recovery anchor (e.g. +4% $\rightarrow$ +6%). Small green ticks silent.

E) True REVERSAL

When: (1) original move erased (UP: live $\leq$ ref), (2) opposite thr hits (shortest opposite window). Old $\rightarrow$ REVERSED + reverse research push; new opposite episode START is silent (AFTER_REVERSAL). One reverse notification only.

F) Never push

- HOLDING; ordinary WEAKENING (25–59%)
- EXPIRY, market close, manual end
- Second START from longer co-qualifying window
- Same-direction restart while disarmed / still elevated
- Stale START research; wrong episodeld; duplicate idempotency key

A8. Complete SNDK story (accel + fade + re-accel)

5m thr = 3%, reference frozen at \$100 at START.

Time	Price	Move	State / alert
10:00	\$103.10	+3.1%	START $\rightarrow$ research push
10:01	\$103.20	+3.2%	HOLDING — no push
10:15	\$105.20	+5.2%	ACCEL push (+2.1 pp)
10:30	\$105.00	+5.0%	HOLDING
10:45	\$103.75	+3.75%	WEAKENING — no push
11:00	\$102.08	+2.08%	STRONG GIVEBACK; anchor $\sim$ +2%
11:05	\$102.50	+2.5%	Silent small recovery
11:20	\$104.10	+4.1%	RE-ACCEL push
11:40	\$106.00	+6.0%	Further ACCEL if +2 pp

A9. Expiry, global re-arm, 3h→24h anti-repackage

If no new extreme of ≥ 0.5 pp for **60 minutes**, episode EXPIRES silently — even if still +10% from ref. Intent: episode = active momentum, not residual elevation.

Critical spam case leadership cares about:

Time	Market	System must
10:00	SNDK +7% / 3h (24h also hot)	START shortest (3h) + push once
Later	Episode EXPIRES	Disarm UP globally (not only 3h)
15:00	3h cold; 24h still +6%	NO START / NO "+6% in 24h" push
Many polls	24h remains qualified	Still no new UP episode
Cool-off	Max UP across windows < re-arm	Direction ARMED
Fresh	New 5m +3.1%	NEW START allowed

Engineering: terminal → markDirectionDisarmed (ticker+direction). Re-arm / START gate uses maxAbsMoveInDirection across all ≤ 24 h/day windows — not only the candidate window — so a 5m thr hit cannot restart while 24h is still elevated. Re-arm level = start window thr – 1 pp. Covered by unit test: *cross-window: expired 3h UP must not re-START on still-hot 24h*.

A10. Reversal, market close, races & concurrency

True reverse: erase + opposite thr → REVERSED + silent opposite START on shortest opposite window.

Market close (equities): US cash CLOSED → CLOSED_AT_MARKET_CLOSE silent. Commodities/crypto/forex do not hard-close the same way on US equity close.

Race	Behaviour
Research #001 finishes after #002 live	No push for #001
Expire mid-research	No START push
Research > ~5 min late	Research saved; push suppressed
Two workers START same second	claimActiveEpisode — one wins
Push retry	idempotencyKey = episodeId:eventType:cycle
DB invariant	Partial unique index: one ACTIVE per ticker

A11. Episode alert cheat-sheet & FAQ

Push YES

- ✓ START — research done, ACTIVE, trigger $\leq$ ~5 min
- ✓ Acceleration — $\geq$ +2 pp beyond last notified
- ✓ Strong giveback — $\geq$ 60% fade, once per cycle
- ✓ Re-acceleration — $\geq$ +2 pp from recovery anchor
- ✓ True reverse — erase + opposite thr (one reverse push)

Push NO

- ✗ Holding / ordinary Weakening
- ✗ Expire / market close / manual end
- ✗ Second START from longer window; disarmed restart; stale research; noise ticks

Boss FAQ

Q: Why not alert every 1% tick?

A: Noise. Only milestones.

Q: Why expire if still +10%?

A: Active momentum, not residual elevation; re-arm stops re-alerting the same move.

Q: Why shortest window?

A: Closest to what just happened; avoids "+8% in 24h" packaging of a 5m trigger.

Q: Weakening vs Reversal?

A: Weakening = faded from peak, still same side of ref. Reversal = through ref + opposite thr.

Q: Sandisk-only?

A: No — same engine for every watchlist ticker; SNDK is the teaching example.

Q: Active episodes per ticker?

A: At most one (memory claim + DB unique ACTIVE).

Episode mental model: poll → windows → no ACTIVE? re-arm + shortest thr → freeze ref → research → START push if fresh → ACTIVE? always vs frozen ref → +2pp / 25% / 60% / reverse / 60m expire → terminal → re-arm only after cool-off across all windows.

Part B — Daily Digest

B1. What it is (vs episodes)

Daily Digest is a **scheduled session pulse** for equities and indexes. It is **not** the episode state machine. It does not create, modify, expire, reverse, or re-arm episodes. Code: `server/momentum/dailyDigest.js`.

	Episode	Daily Digest
When	Threshold cross (5m/3h/...)	Fixed clock: open+10m, midday, close+5m
Measure	Frozen window reference	Always vs previous session close
Per day	0–many (momentum-driven)	Max 3 per ticker (OPEN/MIDDAY/CLOSE)
State machine	STARTED → ... → EXPIRE/REVERSE	Slot rows only
Touches episodes?	Yes	Never
Example push	SNDK surging +3.2% in 5 min	SNDK opens strong ↑ +1.4%
UI rail	Recent events → Episodes	Recent events → Daily Digest

Boss line: Episodes = “is momentum active right now?” Digest = “three fixed check-ins vs yesterday’s close.” Same ticker, different clocks, different math, different rails.

B2. Schedule, eligibility, engineering flow

Timezone: **America/New_York**. Default RTH 09:30–16:00 ET.

Slot	Fire time (ET)	Meaning
OPEN	09:40 (open + 10 min)	Opening move settled
MIDDAY	~12:45 (session midpoint)	Half-day snapshot
CLOSE	16:05 (close + 5 min)	Day-end summary

- Eligible: equities + indexes only (SNDK, AAPL, SPY...). Skip crypto / forex / commodities.
- Skip weekends + known US market holidays.
- Disable: `MOMENTUM_DAILY_DIGEST_ENABLED=0`.
- Dedup key: (ticker, session_date, slot) in Supabase + in-memory claim — one push per slot.
- Flow: due slots → quote → day% vs prev close → claim → research → Expo push (watchlist-gated) → finalize.
- Storage: `momentum_daily_digests`; hydrate into `status.digests` for UI.
- Cycle: `runDailyDigestCycle` over watched tickers; does not touch episodes.

B3. Push copy examples & body structure

Slot	Example title (SNDK)	Measure
OPEN	■ SNDK opens strong ↑ +1.4%	Day vs prev close
OPEN	■ SNDK opens lower ↓ 2.0%	Day vs prev close
MIDDAY	■ SNDK holds higher ↑ +2.8% at midday	Day vs prev close
CLOSE	■ SNDK closes higher ↑ +1.9% today	Day vs prev close

Body = lead line (session position vs prior close) + research likely-driver text (or “No clear catalyst...” if research fails). Event type: `DAILY_DIGEST_OPEN` | `MIDDAY` | `CLOSE`. Push path: same watchlist-gated Expo path as episode alerts (`sendTriggerEpisodePush`, `appKey` trigger).

Part C — One full SNDK trading day

Assumptions: Tuesday US regular session; SNDK on Trigger watchlist; 5m thr $\geq 3\%$; previous close **\$100.00**. Digests always use day vs prev close. Episodes use frozen short-window ref.

C1. Interleaved timeline (America/New_York)

Time	Channel	What happens	Episode impact
09:30	—	Market open	No episode yet
09:35	EPISODE	START #041 UP (5m +3.4%); research push	ACTIVE STARTED→HOLDING
09:40	DIGEST	OPEN: day +3.1% vs \$100	None — independent
09:55	EPISODE	ACCEL +2.1 pp (same frozen ref)	ACCELERATING
10:20	EPISODE	Pullback ~30% giveback — no push	WEAKENING
10:45	EPISODE	STRONG GIVEBACK 60% push + anchor	STRONGLY_WEAKENING
11:10	EPISODE	RE-ACCEL +2 pp vs anchor	RE_ACCELERATING
11:30–12:40	EPISODE	Quiet near peak	HOLDING
12:45	DIGEST	MIDDAY: day +2.8% vs \$100	None
~13:50	EPISODE	EXPIRE silent (60m no ≥ 0.5 pp extreme)	#041 EXPIRED; UP disarmed
14:55	EPISODE	Still elevated day/24h — no new START	Disarmed holds
15:40	EPISODE	Optional: cool + fresh 5m thr → START #042	New episode if re-armed
16:00	EPISODE	If still ACTIVE → silent MARKET CLOSE	CLOSED_AT_MARKET_CLOSE
16:05	DIGEST	CLOSE: day +1.9% vs \$100	None

ASCII view of the two channels the same day:

09:35 EPISODE START · 09:40 DIGEST OPEN · 09:55 ACCEL · 10:45 STRONG FADE · 11:10 RE-ACCEL · 12:45 DIGEST MIDDAY · ~13:50 EXPIRE (silent) · 15:40 optional new START · 16:00 archive · 16:05 DIGEST CLOSE

C2. Alert count & boss talking points

#	Kind	Approx time
1	Episode START	09:35
2	Digest OPEN	09:40
3	Episode ACCEL	09:55
4	Episode strong giveback	10:45
5	Episode re-accel	11:10
6	Digest MIDDAY	12:45
—	Expire (no push)	~13:50
7	Episode START #042 (optional)	~15:40
8	Digest CLOSE	16:05

- **Digest always 0–3/day** per equity (OPEN/MIDDAY/CLOSE once). Episode alerts are variable (milestones only).
- **09:40 OPEN and 09:35 START can both say “up”** — different math: short frozen window vs day vs prev close.
- **Episode expire does not cancel** MIDDAY or CLOSE digests.
- **Digest never re-opens or re-arms** an episode.
- **CLOSE at 16:05** can fire even if the episode was archived at 16:00.

Final wrap for leadership:

- **Momentum channel (episodes)** — is an active directional move underway? Shortest window, frozen ref, milestone pushes, expire when momentum goes quiet, global re-arm so we do not re-package the same move as a longer window later.
- **Calendar channel (daily digest)** — three fixed session check-ins vs previous close for equities/indexes, research-backed, watchlist-gated, independent of the episode state machine.
- Together they answer different questions on the same ticker without spamming every price tick.

— End of brief — · NewsLabs Trigger · Example ticker SNDK · Episodes + Daily Digest